

Prevalent Social Risk Without Corresponding Differences in Blood Pressure Control in an Integrated Health System

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Introduction

The relationship between social risk (e.g., housing instability, food insecurity) and hypertension (HTN) control is poorly understood within integrated healthcare networks.

Objective/Research Question

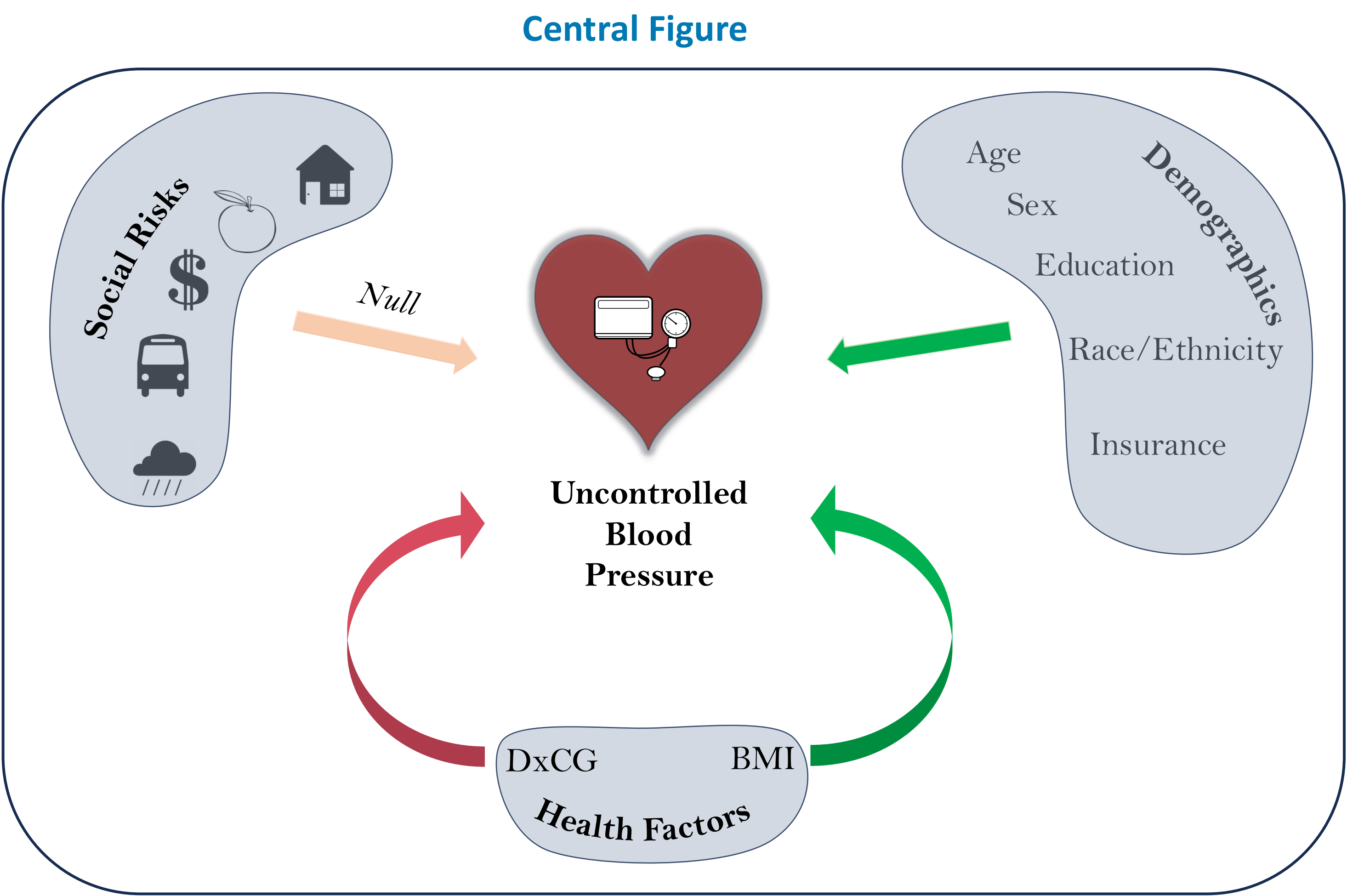
We examined the interplay between social risk and HTN control to identify drivers of uncontrolled HTN and inform strategies to enhance BP control and reduce disparities.

Methods

- Retrospective observational cohort study within KP’s National Social Needs Survey, linking self-reported social risk exposures with HTN control status (HEDIS metrics).
- Eligible respondents (*N* = 686) were aged 18–85 with ≥2 HTN diagnoses and continuous enrollment during 2020.
- The primary outcome was blood pressure (BP) control (Y/N).
- Covariates included age, sex, race/ethnicity, insurance, employment, education, diagnostic cost group (DxCg), BMI, and neighborhood deprivation index.
- Statistical analysis included bivariate comparison followed by multivariable logistic regression.

Conclusions

Social risk was *not* associated with BP control in an integrated healthcare network, perhaps reflecting mitigation of disparity through integrated care and KP’s HTN quality improvement programs.



Central Figure. Social risk, demographics, and health factors exert differential effects on patient risk of having uncontrolled blood pressure.

Results

- Among 686 patients, 36% had uncontrolled HTN and 59% reported social risk (severe risk: 29%).
- Uncontrolled BP was lowest in patients >65 years (*p*<.001) and Medicare/dual enrollees (*p*<.001), and highest in low DxCg (*p*<.001).
- Bivariate analyses showed no difference in BP control by any vs. no social risk (*p*=.129) or by individual risk type (*p*≥.067).
- In multivariable models, older age (OR 0.15 [0.05, 0.49]) was associated with lower odds of uncontrolled HTN; low DxCg (OR 6.63 [2.85, 15.4]) and Medicare/dual insurance (OR 2.75 [1.05, 7.15]) with higher odds.
- Social risk was not associated with HTN control in any model (*p*≥.145).

Stepwise Multivariable Model

Characteristics	Response Options	Hypertension Control		
		OR* (95% CI) [†]		
Enabling Factors		Model 1, Unadjusted Model: Social Risk Only	Model 2, Adjusted Model: + Patient Factors	Model 3, Adjusted Model: + Health Status
Any Social Risk	None (ref)	—	—	—
	Any risk	1.56 (0.98, 2.49)	1.45 (0.89, 2.36)	1.61 (0.94, 2.77)
Demographics				
Age Group	25–49 years (ref)		—	—
	50–64 years		1.03 (0.49, 2.16)	0.87 (0.39, 1.94)
	65+ years		0.24 (0.08, 0.71)	0.15 (0.05, 0.49)
Sex	Female (ref)		—	—
	Male		1.41 (0.89, 2.24)	1.55 (0.92, 2.61)
Race & Ethnicity	White (ref)		—	—
	African American		1.29 (0.61, 2.71)	1.47 (0.66, 3.26)
	Asian		0.89 (0.46, 1.72)	0.80 (0.41, 1.54)
	Hispanic		0.77 (0.39, 1.50)	0.77 (0.36, 1.64)
	Multi-racial		1.13 (0.34, 3.73)	1.33 (0.33, 5.32)
Insurance Type [‡]	Commercial (ref)		—	—
	Individual		0.62 (0.18, 2.11)	0.60 (0.16, 2.29)
	Medicare/Dual		1.53 (0.65, 3.61)	2.75 (1.05, 7.15)
	Medicaid		1.62 (0.52, 5.10)	2.07 (0.59, 7.35)
Education Level	4-year degree or more (ref)		—	—
	Some college or 2-year degree		0.86 (0.50, 1.50)	0.86 (0.47, 1.56)
	High School GED or less		0.60 (0.33, 1.12)	0.59 (0.30, 1.17)
Health Status				
BMI	Healthy (ref)			—
	Underweight			3.19 (0.08, 125)
	Overweight			0.50 (0.25, 1.01)
	Obesity			0.59 (0.28, 1.22)
DxCG	Very High >3 (ref)			—
	High 1.3–3			1.50 (0.80, 2.81)
	Moderate (0.7–1.3)			1.08 (0.50, 2.35)
	Low or Very Low (<0.7)			6.63 (2.85, 15.4)

Central Table. *OR = odds ratio of having uncontrolled blood pressure; [†]CI = confidence interval; [‡]UCDA’s simplified product or insurance type category. Odds ratios >1 indicate higher odds of uncontrolled hypertension. Emboldened values demonstrate statistical significance.